

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q23-2364

Glen Oaks 138 kV  
January 03, 2025

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2364. The Interconnection Customer proposes a 227.2 MW solar plus storage facility interconnecting at the Glen Oaks 138 kV substation in the Virginia area. The request was received on December 07, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 538 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,240,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

The TANGY-BARKSDALE 138 kV is overloaded to 108.5% of its 214 MVA emergency rating for the loss of the BEAR-CLOVER 345 kV. The project contributes 6.2% DFAX (14.1 MW) to this overload.

NEW LONDON-THORNHILL SUBSTATION 138 kV: 108.4% loading (242 MVA) following the outage of the LA JOIE-CARSON ICE COGEN 138 kV line; DFAX = 7.0%.

The THORNHILL-CARSON ICE COGEN 138 kV L is overloaded to 104.5% of its 234 MVA emergency rating for the loss of the LA JOIE-CARSON ICE COGEN 138 kV. The project contributes 7.0% DFAX (15.9 MW) to this overload.

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                          | Required Upgrade           | Cost Estimate | Allocation to Project |
|-----------------------------------|----------------------------|---------------|-----------------------|
| TANGY-BARKSDALE 138 kV            | Reconductor line           | \$13,131,000  | 46% (\$6,029,000)     |
| NEW LONDON-THORNHILL 138 kV       | Reconductor line           | \$5,588,000   | 28% (\$1,557,000)     |
| THORNHILL-CARSON ICE COGEN 138 kV | Reconductor line           | \$2,348,000   | 28% (\$654,000)       |
|                                   | Total allocated to project |               | \$8,240,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.